

Pest Detection Agent - Diagram, Pipeline and Useful URLs

Architecture Diagram

```
graph TD
    Farmer --> WebApp[Web Application]
    WebApp --> Flask[Python Flask]
    Flask --> AIModel[AI Model TensorFlow/Keras]
    AIModel --> MySQL[MySQL]
    MySQL --> Prediction[Prediction Result]
```

Pipeline

```
graph TD
    A[Collect dataset] --> B[Preprocess images]
    B --> C[Train CNN model]
    C --> D[Save model .h5]
    D --> E[Build Flask backend]
    E --> F[Create frontend]
    F --> G[Predict uploaded image]
    G --> H[Store history]
    H --> I[Display result]
```

Workflow

```
graph TD
    Upload[Upload Image] --> Preprocess[Preprocess]
    Preprocess --> AIModel[AI Model]
    AIModel --> DetectPest[Detect Pest]
    DetectPest --> Treatment[Treatment Suggestion]
    Treatment --> Database[(Database)]
    Database --> Result[Result]
```

Useful URLs

<https://code.visualstudio.com/>

<https://www.python.org/>

<https://flask.palletsprojects.com/>

<https://www.tensorflow.org/>

<https://keras.io/>

<https://opencv.org/>

<https://dev.mysql.com/downloads/workbench/>

<https://www.kaggle.com/datasets>